# Witbank Elite Car Wash

INT316D — Web Development Group Project

Semester 1, 2026

Individual Reflection Statements

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## Part A: Individual Reflection Statements

The following five reflections represent the five core roles in the development team: UI/UX Designer, Backend Developer, Frontend Developer, Database Manager, and QA / Tester. Each reflection is approximately 1000 words and addresses: the student's specific technical contributions, collaboration and group dynamics, lessons learned, and outlook.

### Reflection Statement

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**Role:** UI/UX Designer

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#### Introduction

This semester has been a transformative and humbling experience in my growth as a web technologist. Joining the Witbank Elite Car Wash project gave me an extraordinary opportunity to translate creative ideas into a living, functional digital product that serves a real Witbank business. As the UI/UX Designer within our team of ten, my primary responsibility was to shape every visual interaction the user has with the application — from the moment they land on the homepage to the instant they receive a booking confirmation. This was not simply about making screens look attractive; it was about engineering trust, guiding behaviour, and communicating the premium identity of Witbank Elite through every pixel and animation. This reflection documents my design journey, the technical decisions I made, the collaborative friction I navigated, and the lasting lessons I carry into my professional future.

## Technical Contribution and Implementation

My core contribution was the design system and frontend implementation of the glassmorphism UI. Working within the constraints of Vanilla CSS3 — no Bootstrap, no Tailwind — I developed a comprehensive set of CSS custom properties (variables) that formed the backbone of our visual language. Variables like --glass-bg, --neon-cyan, and --border-glass ensured visual consistency across all pages without introducing a dependency on external frameworks, which aligned perfectly with the project's philosophy of lean, modern development.

The glassmorphism aesthetic required careful engineering. The backdrop-filter: blur() CSS property, which creates the frosted-glass look on our service cards and navigation bar, is computationally expensive on lower-end devices. I had to test extensively on multiple screen sizes and browser configurations to find the right balance between visual richness and rendering performance. I ultimately settled on blur(20px) for primary containers and blur(10px) for smaller interactive elements, a decision that preserved the premium feel without causing jank on midrange Android devices.

Implementing the responsive navigation — what we termed the 'glass-nav' — was among the most technically demanding tasks. Using CSS Grid for the overall layout and Flexbox for component-level alignment, I ensured the navigation adapted gracefully from a 320px mobile screen to a 4K desktop. I also introduced micro-animations: subtle CSS transitions on hover states for buttons and cards using cubic-bezier easing curves, giving the interface a sense of tactile physicality that distinguished our product from generic academic web projects.

The typography system was another considered decision. I selected the Outfit font family from Google Fonts for its clean geometric letterforms, which resonate with the automotive and luxury industries. Establishing a clear typographic hierarchy — with distinct sizes and weights for headings, subheadings, body copy, and labels — meant users could scan the page effortlessly and reach the booking form with minimal cognitive load.

## Collaboration and Group Dynamics

Working within a group of ten students across different technical disciplines was the most socially complex aspect of the project. As the UI/UX Designer, I occupied an unusual boundary role: I needed to communicate with the Frontend Developer about HTML structure, with the Backend Developer about Thymeleaf template integration, and with the Project Lead about feature prioritisation — all while maintaining a coherent design vision.

Early in the project, there was significant disagreement about the colour scheme. Some team members preferred a light, minimalist theme for perceived professionalism, while I advocated for the dark neon palette that ultimately defined our product. I resolved this tension not through assertion but through evidence: I created rapid mockups of both approaches and presented them to the team with reference to the target customer profile of Witbank Elite — a premium, aspirational brand serving clients who expect a luxury experience. The dark theme won the vote decisively. This experience taught me that design decisions carry more weight when grounded in user empathy and business context rather than personal preference.

I also had to adapt my design specifications to the realities of Thymeleaf server-side rendering. Certain dynamic UI patterns I envisioned — like a live service price update when a user selects a package — required close coordination with the Frontend Developer to implement via JavaScript event listeners rather than a React-style reactive approach. This constraint, while initially frustrating, deepened my understanding of how design and implementation are inseparable in real-world software.

## Lessons Learned and Future Outlook

The most profound lesson I learned is that great UI/UX design is invisible — it is the absence of friction rather than the presence of decoration. When a user can book a car wash in under two minutes without reading a manual or consulting a help page, the design has succeeded. Every decision I made — button size, colour contrast ratio, form field spacing, confirmation feedback — was in service of that frictionless experience.

I also discovered the power of CSS custom properties as a design-to-code bridge. By defining the visual language as variables, I created a system that both I and the Frontend Developer could reason about, reducing miscommunication and rework. In the future, I intend to explore design tokens — a more formalised version of this approach used in large-scale design systems at companies like Google and Airbnb.

This project also made me acutely aware of accessibility as a design responsibility, not an afterthought. Our dark glassmorphism aesthetic, while visually striking, presented contrast challenges. Ensuring WCAG AA-compliant contrast ratios between text and glass backgrounds required iterative testing using browser accessibility tools — a practice I will embed in every future project from the outset.

## Conclusion

The Witbank Elite Car Wash project was the most complete creative and technical challenge I have faced in my academic career. Designing a product that a real business could plausibly deploy — one that addresses genuine operational pain points while projecting a premium brand identity — made every late night worthwhile. I am proud of the visual language we created and grateful for a team that trusted the design process. I leave this semester with a far richer understanding of the relationship between aesthetics, engineering, and human behaviour, and I

am eager to carry these foundations into a professional UI/UX career.

## Reflection Statement